Dr. Krishna Kumar Yadav

Postdoctoral Researcher — 2D Materials, Nanofabrication & Device Physics

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RESEARCH EXPERTISE

- **2D Material Synthesis & Transfer:** Graphene, Transition Metal Dichalcogenides ($MoTe_2$, SnS_2), Black Phosphorus, h-BN heterostructures.
- **Cleanroom Nanofabrication & Lithography:** Dip-Pen Nanolithography (DPN), optical lithography, micro/nanofabrication process development.
- **Device Physics & Integration:** Field-Effect Transistors (FETs), optoelectronic devices, engineered metal–semiconductor contacts, planar terahertz (log-periodic / bowtie antenna) detector concepts.
- **Characterization:** Atomic Force Microscopy (multi-mode), Raman spectroscopy, SEM, XRD / HT-XRD, frictional force and transverse shear microscopy.
- **Sensing Technologies:** Electrochemical sensors for heavy-metal detection (Pb, Hg, Sb); field-emission-based sensing.
- **Electrocatalysis & Energy:** Water splitting, supercapacitors, photoelectrochemical systems.

EDUCATION

Ph.D. in Physics — Institute of Nano Science & Technology (INST), Mohali, Punjab, India *(2021)*

Thesis: *Nanostructured Metal Borides: Synthesis and Their Applications*. Supervisors: Dr. Menaka Jha & Prof. Ashok K. Ganguli.

M.Sc. in Physics — Mahatma Gandhi P.G. College, D.D.U. Gorakhpur University, Gorakhpur, India *(2014)*

B.Sc. (Physics, Chemistry, Mathematics) — Marwar Business School, D.D.U. Gorakhpur University, Gorakhpur, India *(2012)*

PROFESSIONAL EXPERIENCE

Postdoctoral Researcher, Department of Fundamental Physics, University of Salamanca, Spain *Sep 2024 – Present*

Postdoctoral Researcher, Shamoon College of Engineering, Beer-Sheva, Israel *Nov 2022 – Aug 2024*

Project: Dip-Pen Nanolithography printing over screen-printed electrodes for real-time electrochemical sensing of heavy metals.

Institute Postdoctoral Fellow, Institute of Nano Science & Technology (INST), Mohali, India *Jun 2021 – Oct 2022*

Project: Understanding the role of 2D materials for field emission.

Senior Research Fellow (SRF), CSIR-India — Ph.D. Research, Institute of Nano Science & Technology (INST), Mohali, India *Jan 2018 – Jun 2021*

Total post-Ph.D. research experience: 5+ years, spanning four research institutions across three countries (India, Israel, Spain).

HONOURS AND AWARDS

- CSIR-NET in Physical Science, Council of Scientific & Industrial Research, India (2015).
- Best Poster Award — “Extraction of Valuable Nanostructured Materials from Industrial Waste,” 9th Bangalore India Nano, 7–9 December 2017.

INVITED TALKS, CONFERENCE PRESENTATIONS & WORKSHOPS

1. Invited Talk: “Optimization of 2D Materials for Advanced Solar Cells Using Engineered Metal Contacts,” 4th Conference on Advanced Materials in Spain, Universitat Autònoma de Barcelona, 24–26 November 2025.

2. Invited Talk: "Optimization of 2D Materials for Advanced Solar Cells Using Engineered Metal Contacts," International Conference on Layered Materials and Devices: From Atoms to Chips!, INL – International Iberian Nanotechnology Laboratory, Braga, Portugal, 4–6 November 2025.
3. Poster: "Tribological Properties of Mono and Few Layers of $MoTe_2$ by Frictional Force and Transverse Shear Microscopy," GEFES 2025, Oviedo, Spain, 29–31 January 2025.
4. Poster: "Excellent Field Emission from Ultrafine Vertically Aligned Nanorods of NdB_6 on Silicon Substrate," ICONSAT 2018, Bengaluru, India.
5. Poster: "Extraction of Valuable Nanostructured Materials from Industrial Waste," 9th Bangalore India Nano, 7–9 December 2017.
6. Workshop: "X'Radiate – Workshop on Powder, Nano and Thin Film Characterization Using X-Ray Diffraction," PSGTECHS COE INDUTECH with PANalytical India, 15–16 December 2016.
7. Participant: 10th CRSI-RSC Joint Symposium and 18th CRSI National Symposium in Chemistry, Institute of Nano Science & Technology and Punjab University, 4–7 February 2016.

RESEARCH MENTORING

- Rubén Bastida Vadillo (2025–2026, TFG, Spain) — Co-supervised with Ana Pérez Rodríguez. "Fabrication and Characterization of $MoTe_2$/$NbSe_2$ Heterostructures for High-Performance Photodetectors," NanoLab, Dept. of Fundamental Physics, University of Salamanca, Spain.
- Alex Bellido Escribano (2025–2026, TFG, Spain) — Co-supervised with Mario Amado Montero. "New Techniques for Obtaining Ultrapure Graphene," NanoLab, Dept. of Fundamental Physics, University of Salamanca, Spain.
- Ahmed Belal Salik Usmani (Aug 2021–May 2022, M.Tech Internship) — with Dr. Menaka Jha. "Supercapacitive Performance of Graphene Derived from Waste Pine Tree Leaves," INST, Mohali, India.
- Supriya Rana (2017–2020, Ph.D.) — with Dr. Menaka Jha. "Synthesis of Crystalline Fe_2O_3 and Energy Application," INST, Mohali, India.
- Nausad Khan (2018–2020, Ph.D.) — with Dr. Menaka Jha. "Green Route Synthesis of MoS_2 over Graphite for Energy Application," INST, Mohali, India.
- Kritika Sood (2018–2021, Ph.D.) — with Dr. Menaka Jha. "Valuable Materials from Tata Steel Slime," INST, Mohali, India.
- Ritika Wadhwa (2018–2021, Ph.D.) — with Dr. Menaka Jha. "Black Gold Decorated Monoclinic Zirconia Synthesis," INST, Mohali, India.
- Shubham Vishnoi (Mar–Jul 2020, M.Sc. Internship) — with Dr. Menaka Jha. "Technologies for Arsenic Removal from Water," INST, Mohali, India.
- Priyanka Goyal (May–Jun 2016, Graduate Internship) — with Dr. Menaka Jha. "Synthesis of Cobalt Ferrite and Its Application as Electrocatalyst," INST, Mohali, India.
- Henna Sammi (May–Jun 2016) — with Dr. Menaka Jha. "Development of a Prototype for the Removal of Toxic Impurities from Water," INST, Mohali, India.

COMMUNITY OUTREACH

- Demonstration of INST technologies (water purification and valuable-materials extraction from industrial waste) at INST, Mohali, on National Technology Day, 11 May 2018.
- Demonstration on "Valuable Materials Extraction from Battery Waste" at Bangalore Nano, Bangalore, India, 7–8 December 2017.
- Demonstrated the "Water Purification Technique" and "Waste-to-Valuable-Materials Extraction" at the 3rd India International Science Festival, IIT Madras, Chennai, 13–16 October 2017.
- Demonstration at CSIR, Delhi, under the "Swachhata Hi Seva" programme by the Government of India, 28 September – 2 October 2017.
- Demonstration at "Destination Himachal Pradesh 2017," Palampur, India, 12–14 September 2017.

HANDS-ON RESEARCH INSTRUMENTATION

- 2D-material transfer system; Dip-Pen Nanolithography (DPN), optical lithography, wire bonding.

- Homemade microscope for 2D-flake thickness measurement.
- Homemade field-emission setup; Atomic Force Microscopy (multiple modes).
- Raman spectrometer (WITEC); Scanning Electron Microscope (JEOL JSM-IT300).
- X-ray Diffraction and High-Temperature XRD up to 1200 °C (Bruker D8 Advance).
- Electrochemical Workstation (Metrohm PGSTAT-30 / CH Instruments / PalmSens).
- Thermogravimetric Analysis (Perkin Elmer STA-8000); Differential Scanning Calorimetry (Perkin Elmer DSC-8000).
- Surface Area measurement (Quantachrome Autosorb iQ2).

PEER-REVIEWED PUBLICATIONS (51)

Manuscripts under review / submitted:

- Krishna K. Yadav, et al. Room-Temperature Photodetector Performance of Monolayer 2H-$MoTe_2$: Effects of Trap Density and Contact Barrier. Submitted to Small.
- Krishna K. Yadav, et al. Robust Sliding-Induced Electrostatic Domain Patterns in Exfoliated ReS_2 Flakes. Submitted to 2D Materials.

Complete list:

1. Shape-induced manganese iron-based nanostructures loaded onto iron sheet as efficient electrocatalyst for O_2 evolution. Inorganic Chemistry Communications, 2026, 183, 115745.
2. Sujit Kumar Guchhait, Supriya Rana, Krishna K. Yadav, Sushma Kumari, Surinder K. Mehta, Menaka Jha. Excellent Electrochemical Oxygen Production from Ultrafine Nanoparticles of Manganese Ferrite Spinel Derived from Spent Primary Battery. Energy Technology, 2025, 13, 2500985.
3. Ritika Wadhwa, Arushi Arora, Sushma Kumari, Krishna K. Yadav, Menaka Jha. Enhanced Photogenerated Carrier Assisted Ethanol Oxidation via Surface Modification with Perovskite to Develop p-n type NiTiO3-NiO Heterostructure. Materials Research Bulletin, 2026, 196, 113865.
4. Krishna K. Yadav, Dror Shamir, Haya Kornweitz, Lonia Friedlander, Moshe Zohar, Ariela Burg. Electrochemical Sensor Based on Black Phosphorus for Antimony Detection Using Dip-Pen Nanolithography: The Role of Dwell Time. Small Methods, 2025, 2402157. (IF~9.1)
5. Krishna K. Yadav, Sunaina, S. Saini, R. Wadhwa, S. Devi, S. Ghosh, M. Jha. Black Phosphorus/Dysprosium Hexaboride-Based Heterostructured Films for Field Emission Technologies. ACS Applied Nano Materials, 2024, 7, 9942–9949. (IF~5.5)
6. Krishna K. Yadav, R. Wadhwa, V.M. Gaikwad. Er2FeCrO6: Emerging Efficient Nanomaterials for Multiferroics. Materials Chemistry and Physics, 2024, 317, 129198. (IF~4.7)
7. Krishna K. Yadav, Dror Shamir, Haya Kornweitz, Yael Peled, Moshe Zohar, Ariela Burg. Development of Meta-Chemical Surface by Dip-Pen Nanolithography for Precise Electrochemical Lead Sensing. Small Methods, 2024, 8, 1–9. (IF~9.1)
8. Krishna K. Yadav, G. Kumar, S. Rana, S. Ghosh, M. Jha. Understanding the Role of Sheet Thickness on Field Emission from Engineered Hexagonal Tin Disulphide. Applied Surface Science, 2022, 606, 154816. (IF~6.9)
9. Krishna K. Yadav, Gulshan Kumar, Ankush, S. Ghosh, Menaka Jha. Design of a New Process for Stabilization of LaxGd1-xB6 Nanorods and Their Field Emission Properties. Materials Science and Engineering: B, 2022, 282, 115759. (IF~4.6)
10. Krishna K. Yadav, Ankush, Gulshan Kumar, Arushi Arora, S. Ghosh, Menaka Jha. An Insight of Enhanced Field Emission from Anisotropic LaxNd1-xB6 Nanostructures. Materials Chemistry and Physics, 2022, 279, 125694. (IF~4.7)
11. Krishna K. Yadav, Ritika Wadhwa, Nausad Khan, Menaka Jha. Efficient Metal-Free Supercapacitor Based on Graphene Oxide Derived from Waste Rice. Current Research in Green and Sustainable Chemistry, 2021, 4, 100075.
12. Krishna K. Yadav, Harish Singh, Supriya Rana, Sunaina, Heena Sammi, S.T. Nishanthi, Menaka Jha. Utilization of Waste Coir Fibre Architecture to Synthesize Porous Graphene Oxide and Their Derivatives: An Efficient Energy Storage Material. Journal of Cleaner Production, 2020, 276, 124240. (IF~10)
13. Krishna K. Yadav, M. Sreekanth, S. Ghosh, Ashok K. Ganguli, Menaka Jha. Excellent Field Emission from Ultrafine Vertically Aligned Nanorods of NdB_6 on Silicon Substrate. Applied Surface Science, 2020, 526, 146652. (IF~6.9)
14. Krishna K. Yadav, M. Sharma, Menaka Jha. Extraction of Nanostructured Sodium Nitrate from Industrial Effluent and Their Thermal Properties. Water Environment Research, 2020, 92, 1123–1130. (IF~1.9)

15. Krishna K. Yadav, M. Sreekanth, Ankush, S. Ghosh, Menaka Jha. A New Process for Stabilization of Vertically Aligned GdB6 Nanorods and Their Field Emission Properties. CrystEngComm, 2020, 22, 5473–5480. (IF~2.6)
16. Krishna K. Yadav, Sujit Kumar Guchhait, C.M. Hussain, Ashok K. Ganguli, Menaka Jha. Synthesis of Zirconium Diboride and Its Application in the Protection of Stainless-Steel Surfaces in Harsh Environment. Journal of Solid-State Electrochemistry, 2019, 23, 3243–3253. (IF~2.6)
17. Krishna K. Yadav, A. Gupta, M. Sharma, N. Dabas, A.K. Ganguli, M. Jha. Low-Temperature Synthesis Process of Stabilization of Cubic Yttria Stabilized Zirconia Spindles: An Important High Temperature Ceramic Material. Materials Research Express, 2017, 4, 105044. (IF~2.2)
18. Rahma Okbi, Mohammed Alkrenawi, Krishna K. Yadav, Dror Shamir, Haya Kornweitz, Yael Peled, Moshe Zohar, Ariela Burg. Dip-Pen Nanolithography-Based Fabrication of Meta-Chemical Surface for Heavy Metal Detection: Role of Poly-Methyl Methacrylate in Sensor Sensitivity. Small Science, 2025, 5, 2400459. (IF~8.3)
19. Zorik Shamish, Krishna K. Yadav, Haya Kornweitz, Moshe Zohar, Dror Shamir, Raz Jelinek, Ariela Burg. Harnessing the Potential of Dip-Pen Nanolithography to Pattern Meta-Chemical Surfaces with Glutathione Inks: An Electrochemical Sensor for Pb(II) and Hg(II). Advanced Materials Interfaces, 2025, 2500295. (IF~4.4)
20. S. Rana, S.K. Guchhait, Krishna K. Yadav, S. Devi, S.K. Mehta, M. Jha. Hierarchical Nanostructure Engineering of 3D Nickel Cobaltite Ultrafine Nanoparticles Assemblies for Synergistic Electrocatalytic Water and Urea Oxidation. Journal of Electroanalytical Chemistry, 2024, 966, 118378. (IF~4.1)
21. A. Burg, Krishna K. Yadav, D. Meyerstein, H. Kornweitz, D. Shamir, Y. Albo. Effect of Sol–Gel Silica Matrices on the Chemical Properties of Adsorbed/Entrapped Compounds. Gels, 2024, 10, 441. (IF~5.3)
22. Supriya Rana, Ahmed Belal Salik Usmani, Sapna Devi, Ritika Wadhwa, Krishna K. Yadav, Surinder K. Mehta, Menaka Jha. Tailoring Crystalline Mn5O8 Nanostructures: Kinetic Control and Stabilization for Enhanced Electrocatalytic O2 Evolution. Fuel, 2024, 367, 131485. (IF~7.5)
23. S. Rana, Krishna K. Yadav, S. Devi, S.K. Mehta, M. Jha. Oxalate-Mediated Synthesis of Hybrid Nickel Cobalt-Based Nanostructures for Boosting Water and Urea Electrooxidation Efficiency. Journal of Alloys and Compounds, 2024, 990, 174241. (IF~6.3)
24. S. Aharon, S.G. Patra, Krishna K. Yadav, Moshe Zohar, Dan Meyerstein, Eyal Tzur, Dror Shamir, Yael Albo, Ariela Burg. Photoelectrochemical Water Oxidation Reaction for Coated and Meta-Chemical Surface Electrodes with Na3[Ru2(μ-CO3)4]. International Journal of Hydrogen Energy, 2024, 72, 1058–1068. (IF~8.3)
25. A. Arora, R. Wadhwa, Krishna K. Yadav, Ankush, M. Jha. Enhanced Electrochemical Oxygen Generation from Sillenite Phase of Bismuth Iron Oxide (Bi24Fe2O39) Ultrafine Particles Stabilised at Room Temperature. Journal of Electroanalytical Chemistry, 2024, 958, 118154. (IF~4.1)
26. A.B.S. Usmani, S. Rana, A. Arora, Krishna K. Yadav, H. Sammi, N. Sardana, M. Jha. Electrochemical Oxygen Generation from VO2 Nanoflakes Decorated onto Graphite Sheet. Journal of Alloys and Compounds, 2024, 976, 173058. (IF~6.3)
27. N. Khan, Krishna K. Yadav, R. Wadhwa, M. Jha. Realizing Ultralow Overpotential During Electrochemical Hydrogen Generation Through Photoexcitation of Chromium Disulphide. International Journal of Hydrogen Energy, 2024, 56, 1294–1300. (IF~8.3)
28. Parul Sood, Krishankant, Harshita Bagdwal, Arti Joshi, Krishna K. Yadav, Chandan Bera, Monika Singh. Polyoxometalate-Derived Cu-MoO2 Nanosheets as Electrocatalyst for Enhanced Acidic Water Oxidation. ACS Applied Nano Materials, 2024, 7, 69–76. (IF~5.5)
29. Ritika Wadhwa, Krishna K. Yadav, Ankush, Supriya Rana, Sunaina, Menaka Jha. Cobalt Decorated Graphitic Carbon Nitride Photoanode for Electrochemical Ethanol Oxidation: A Sustainable Way Towards Clean Energy. International Journal of Hydrogen Energy, 2023, 48, 29982–29995. (IF~8.3)
30. S.K. Guchhait, Krishna K. Yadav, A. Yadav, R. Zutshi, N. Singh, M. Jha. Design of New Process to Utilize Stubble Char for Construction of M25 Concrete. Advanced Materials Science and Technology, 2023, 5, 1–13.
31. Anima Mahajan, Nausad Khan, Krishna K. Yadav, Menaka Jha, S. Ghosh. Efficient Field Emission from Ultrafine Nanostructured Lanthanum Sulphide Synthesized by Chemical Route. Applied Surface Science, 2023, 623, 156996. (IF~6.9)
32. Kritika Sood, Krishna K. Yadav, Kuldeep K. Bhasin, Tamal K. Ghosh, Santanu Sarkar, Menaka Jha. Unravelling the Behaviour of Dual Active Sites of La2FeMnO6 for Electrochemical Oxygen Generation in Alkaline Conditions. Journal of Electroanalytical Chemistry, 2023, 938, 117449. (IF~4.1)
33. Sujit K. Guchhait, Krishna K. Yadav, Shyam Khatana, Rajendra K. Saini, Upain K. Arora, Rajiv Satyakam, Ramesh Bajaj, Menaka Jha. Conversion of Gaseous Effluents of Power Plant to Sodium Carbonate: A Value-Added Material for Powder Detergent. Cleaner Waste Systems, 2022, 3, 100042. (IF~3.9)

34. S. Devi, Sunaina, R. Wadhwa, Krishna K. Yadav, Menaka Jha. Understanding the Origin of Ethanol Oxidation from Ultrafine Nickel Manganese Oxide Nanosheets Derived from Spent Alkaline Batteries. Journal of Cleaner Production, 2022, 376, 134147. (IF~10)
35. Supriya Rana, Krishna K. Yadav, Sunaina, Surinder K. Mehta, Menaka Jha. Structurally Engineered Anisotropic Cobalt-Based Nanostructures for Efficient Chlorine and Oxygen Evolution. Advanced Materials Interfaces, 2022, 9, 2200740. (IF~4.4)
36. Nausad Khan, Krishna K. Yadav, Ritika Wadhwa, Ankush, Menaka Jha. One Pot Green Method for Fabrication of MoS_2 Decorated Graphite Rods and Their Application in H_2 Evolution. Materials Letters, 2022, 313, 131784. (IF~2.7)
37. Ankush, Krishna K. Yadav, Sujit Kumar Guchhait, Menaka Jha. Surface Phosphorization of Nickel Oxalate Nanosheets to Stabilize Ultrathin Nickel Cyclo Tetraphosphate Nanosheets for Efficient Hydrogen Generation. Materials Chemistry and Physics, 2021, 139, 111275. (IF~4.7)
38. Supriya Rana, Krishna K. Yadav, Sujit Kumar Guchhait, S.T. Nishanthi, S.K. Mehta, Menaka Jha. Insights of Enhanced Oxygen Evolution Reaction of Nanostructured Cobalt Ferrite Surface. Journal of Materials Science, 2021, 56, 8383–8395. (IF~3.9)
39. Ankush, Krishna K. Yadav, Sujit Kumar Guchhait, Supriya Rana, Menaka Jha. Excellent Hydrogen Generation from Ultrathin Nanosheets of Cobalt Cyclo Tetraphosphate. Materials Science and Engineering: B, 2021, 265, 114983. (IF~4.6)
40. Ritika Wadhwa, Krishna K. Yadav, Tanmay Goswami, Ankush, Sujit Kumar Guchhait, Sunaina, S.T. Nishanthi, Hirendra N. Ghosh, Menaka Jha. Mechanistic Insights for Photoelectrochemical Ethanol Oxidation on Black Gold Decorated Monoclinic Zirconia. ACS Applied Materials & Interfaces, 2021, 13, 9942–9954. (IF~8.2)
41. S.K. Guchhait, H. Sammi, Krishna K. Yadav, S. Rana, M. Jha. New Hydrometallurgical Approach to Obtain Uniform Antiferromagnetic Ferrous Chloride Cubes from Waste Tin Cans. Journal of Materials Science: Materials in Electronics, 2021, 32, 2965–2972. (IF~2.8)
42. Supriya Rana, Krishna K. Yadav, Kritika Sood, Ankush, S.K. Mehta, Menaka Jha. Low Temperature Hydrothermal Method for Synthesis of Fe_2O_3 and Their Oxygen Evolution Performance. Electroanalysis, 2020, 32, 2528–2534. (IF~2.3)
43. Yang Gao, Fangru Xing, Menaka Jha, Krishna K. Yadav, Ranjana Yadav, Avtar S. Matharu. Towards Novel Biocomposites from Unavoidable Food Supply Chain Wastes and Zirconia. ACS Sustainable Chemistry & Engineering, 2020, 8, 14039–14046. (IF~7.3)
44. Sunaina, Krishna K. Yadav, S.K. Guchhait, K. Sood, S.K. Mehta, A.K. Ganguli, Menaka Jha. Mechanistic Insights of Enhanced Photocatalytic Efficiency of SnO_2-SnS_2 Heterostructures Derived from Partial Sulphurization of SnO_2. Separation and Purification Technology, 2020, 242, 116835. (IF~9)
45. H. Singh, Sunaina, Krishna K. Yadav, V.K. Bajpai, Menaka Jha. Tuning the Bandgap of m-ZrO_2 by Incorporation of Copper Nanoparticles into the Visible Region for the Treatment of Organic Pollutants. Materials Research Bulletin, 2020, 123, 110698. (IF~5.7)
46. V.M. Gaikwad, Krishna K. Yadav, Sunaina, S. Chakraverty, S.E. Lofland, K.V. Ramanujachary, S.T. Nishanthi, Ashok K. Ganguli, Menaka Jha. Design of Process for Stabilization of La_2NiMnO_6 Nanorods and Their Magnetic Properties. Journal of Magnetism and Magnetic Materials, 2019, 492, 165652. (IF~3.0)
47. N. Dabas, Krishna K. Yadav, Ashok K. Ganguli, Menaka Jha. New Process for Conversion of Hazardous Industrial Effluent of Ceramic Industry into Nanostructured Sodium Carbonate and Their Application in the Textile Industry. Journal of Environmental Management, 2019, 240, 352–358. (IF~8.4)
48. S.T. Nishanthi, Krishna K. Yadav, A. Baruah, A.K. Ganguli, M. Jha. New Sustainable and Environmentally Friendly Process of Synthesis of Highly Porous Mo_2S_3 Nanoflowers in Cooking Oil and Their Electrochemical Properties. Electrochimica Acta, 2019, 300, 177–185. (IF~5.6)
49. S.T. Nishanthi, A. Baruah, Krishna K. Yadav, D. Sarker, S. Ghosh, A.K. Ganguli, Menaka Jha. New Low Temperature Environmentally Friendly Process for the Synthesis of Tetragonal MoO_2 and Its Field Emission Properties. Applied Surface Science, 2019, 467, 1148–1156. (IF~6.9)
50. Vishwajit M. Gaikwad, Krishna K. Yadav, S.E. Lofland, Kandalam V. Ramanujachary, S. Chakraverty, Ashok K. Ganguli, Menaka Jha. New Low-Temperature Process for Stabilization of Nanostructured La_2NiMnO_6 and Their Magnetic Properties. Journal of Magnetism and Magnetic Materials, 2019, 471, 8–13. (IF~3.0)
51. S.T. Nishanthi, Krishna K. Yadav, A. Baruah, K. Vaghasiya, R.K. Verma, A.K. Ganguli, Menaka Jha. Nanostructured Silver Decorated Hollow Silica and Their Application in the Treatment of Microbial Contaminated Water at Room Temperature. New Journal of Chemistry, 2019, 43, 8993–9001. (IF~2.5)

BOOK CHAPTERS (13)

1. Deepak Kumar Chauhan, Kritika Sood, Venugopala Rao Battula, Krishna K. Yadav. Chemical Waste-Derived Carbon Nanomaterials. ACS Symposium Series, Vol. 1494, 2025, 203–224.
2. Arushi Arora, Krishna K. Yadav. Potentiometric Devices for Biomarkers. Detection Science Series, RSC, 2024, 146–165.
3. Krishna K. Yadav, Sunaina, S. Rana, S.K. Guchhait. Scalable and Cost-Effective Synthesis of 2D Materials. 2D Materials: Sensing Applications, 2024, 1–19.
4. S. Rana, V.M. Gaikwad, Krishna K. Yadav. Quantum Dots/One-Dimensional (1D) Composites. Springer Nature Switzerland, 2024, 177–191.
5. Khushbu Jain, Krishna K. Yadav, Neeru Dabas. Green Synthesis of Nanomaterials and Their Applications for Sustainable Environment. CRC Press, 2024, 19.
6. Krishna K. Yadav, Supriya Rana, Sunaina. Application of Waste-Synthesized Nanoparticles in Energy. Elsevier, 2024, 319–340.
7. Krishna K. Yadav, Sunaina, Menaka Jha. Standard Reference Materials. Elsevier, 2024, 199.
8. R. Wadhwa, A. Arora, Krishna K. Yadav. Advanced Fiber Materials in Optical and Photonic Applications. Walter de Gruyter GmbH & Co KG, 2023, 191.
9. R. Wadhwa, A. Arora, S. Rana, Krishna K. Yadav. Overview of Advanced Fiber Materials. Walter de Gruyter GmbH & Co KG, 2023, 1.
10. Krishna K. Yadav, A. Arora, S. Jangra, M. Jha. Chemically Modified Carbon Nanotubes in 3D and 4D Printing. Wiley-VCH GmbH, 2023, 419–439.
11. Krishna K. Yadav, N. Khan, M. Jha. Application of Surface Modified Carbon Nanotubes in Energy. American Chemical Society, 2022, 101–119.

11 of 13 chapters listed above with full citation details; 2 additional chapters pending full citation confirmation.

PATENTS

1. Santanu Sarkar, Tamal Kanti Ghosh, Supriya Sarkar, Menaka Jha, Sujeet Kumar, Kritika Sood, Krishna Yadav, Ankush. A Process for Producing La_2FeMnO_6 from Iron Ore Slime and Applications Thereof. Patent No. 482517 (Granted).
2. Santanu Sarkar, Niloy Kundu, Tamal Kanti Ghosh, Menaka Jha, Sujit Ghuchait, Krishna Yadav, Sunaina, Arushi Arora. Process of Preparing Silica-Iron Oxide from Iron Ore Slime. Patent No. 577043 (Granted).
3. Menaka Jha, Ritika Wadhwa, Sunaina, Ankush, Krishna Yadav, Sujit Kumar Guchhait. Process of Conversion of Wastepaper to Nanostructured $CaCO_3$ and Their Application in De-fluorination of Contaminated Water. Application No. 202311017356 (Filed).

ACADEMIC REFERENCES

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